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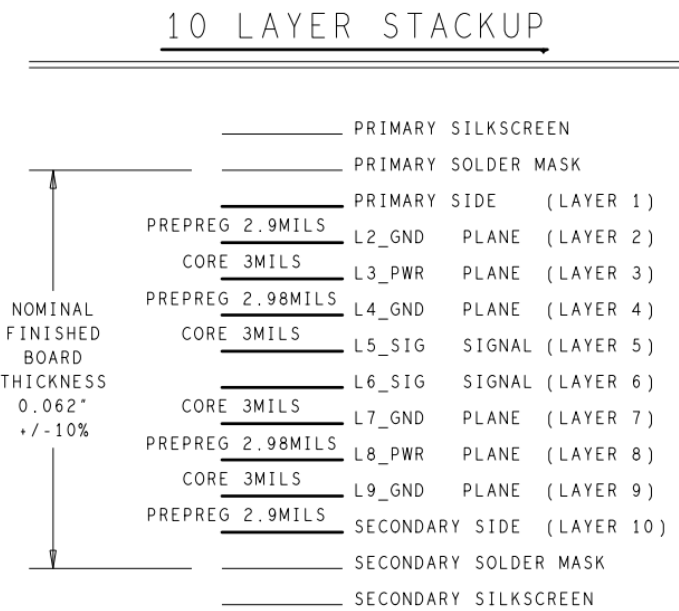
Product Overview

This SDR is a derivative of the official Pluto SDR, which adds gigabit Ethernet ports and TF card interfaces on the basis of retaining ADI's official design.

Technical Details of PCB design

Layered design:

ADI official PCB stack design information:



Stacking information of this product:

板厚	1.6mm	
阻焊颜色	黑色	
字符颜色	白色	
外层铜厚	1OZ	
内层铜厚	0.5OZ	
表面工艺	沉金	
叠层顺序	outline	板框层
1	silk top	
2	mask_top	
3	l1_top	1
4	l2_gnd	2
5	l3_pwr	3
6	l4_gnd	4
7	l5_sig	5
8	l6_sig	6
9	l7_gnd	7
10	l8_pwr	8
11	l9_gnd	9
12	l10_bottom	10
13	mask_bottom	
14	silks_bottom	

To ensure product performance, the same stacked and 10 layer board design is used as ADI official.

Impedance control:

ADI official PCB impedance control information:

IMPEDANCE TABLE

NOTE: ALL IMPEDANCE TOLERANCE +/-10%

SE = SINGLE ENDED, DP = DIFFERENTIAL PAIR

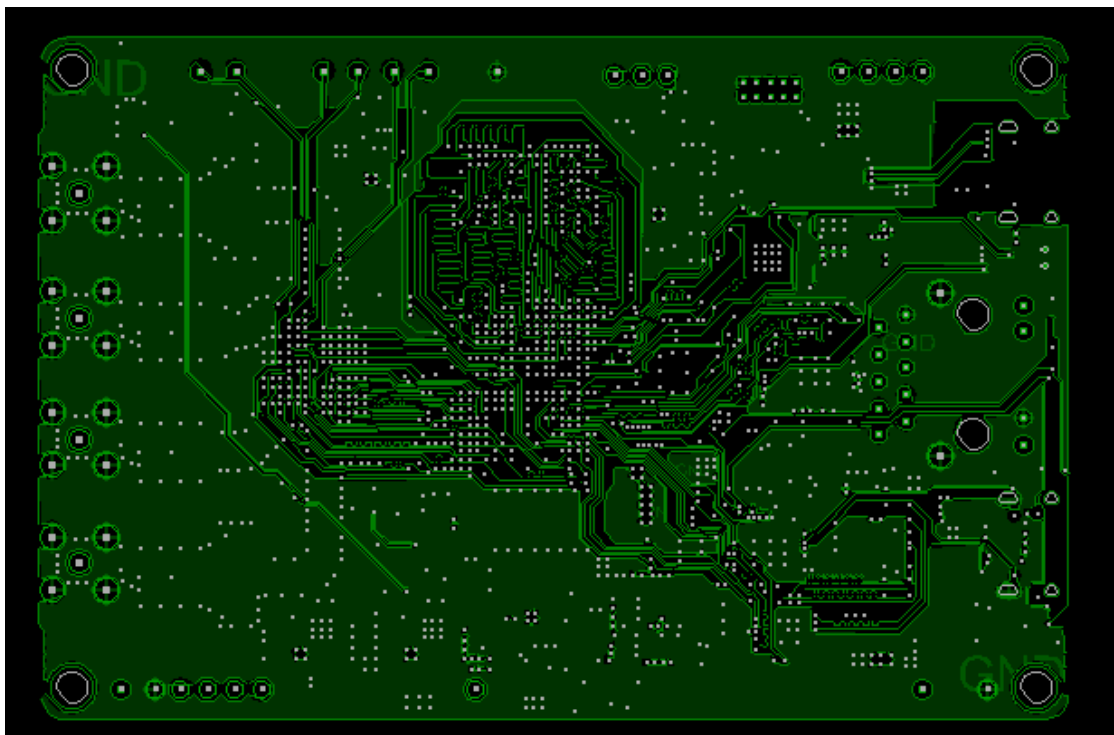
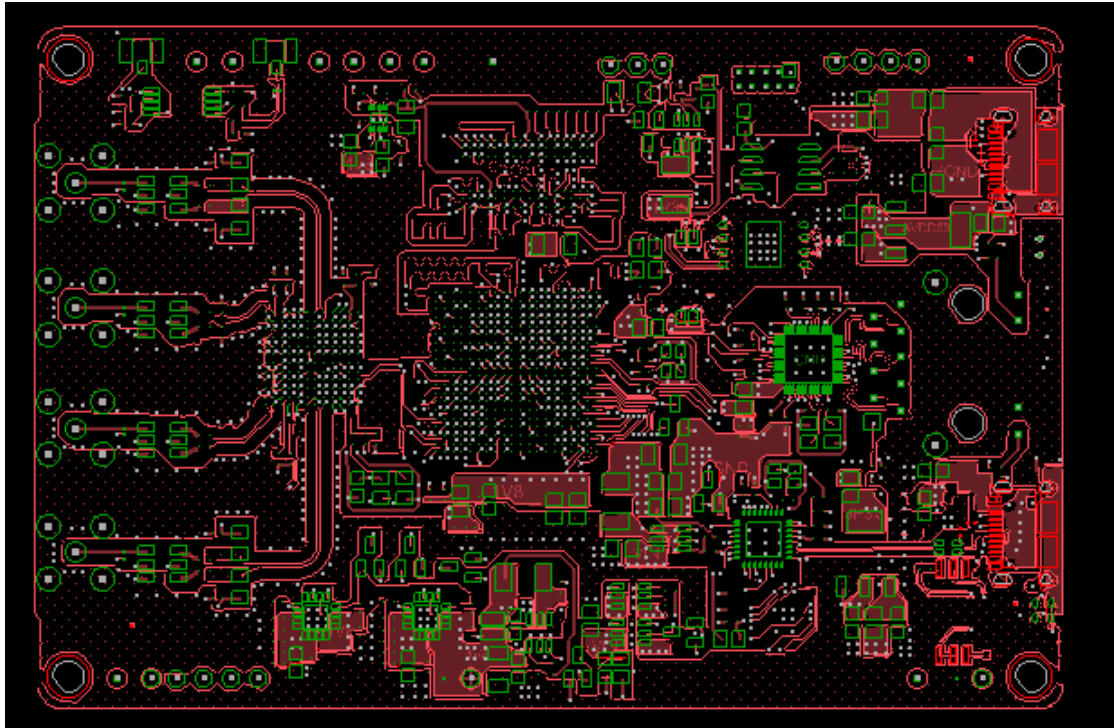
LAYER	IMPEDANCE	REFERENCE	LINE WIDTH	SPACE
L1	50 OHMS SE	L2	4MILS	N/A
L1	50 OHMS SE	L4	19MILS	N/A
L1	40 OHMS SE	L2	6.5MILS	N/A
L1	100 OHMS DP	L4	9MILS	7MILS
L1	90 OHMS DP	L2	4MILS	6.5MILS
L5	40 OHMS SE	L4,L7	6.5MILS	N/A
L5	50 OHMS SE	L4,L7	4.5MILS	N/A
L5	80 OHMS DP	L4,L7	5.5MILS	7MILS
L6	40 OHMS SE	L4,L7	6.5MILS	N/A
L6	50 OHMS SE	L4,L7	4.5MILS	N/A
L6	80 OHMS DP	L4,L7	5.5MILS	7MILS
L10	50 OHMS SE	L9	4MILS	N/A
L10	40 OHMS SE	L9	6.5MILS	N/A
L10	80 OHMS DP	L9	5MILS	6MILS

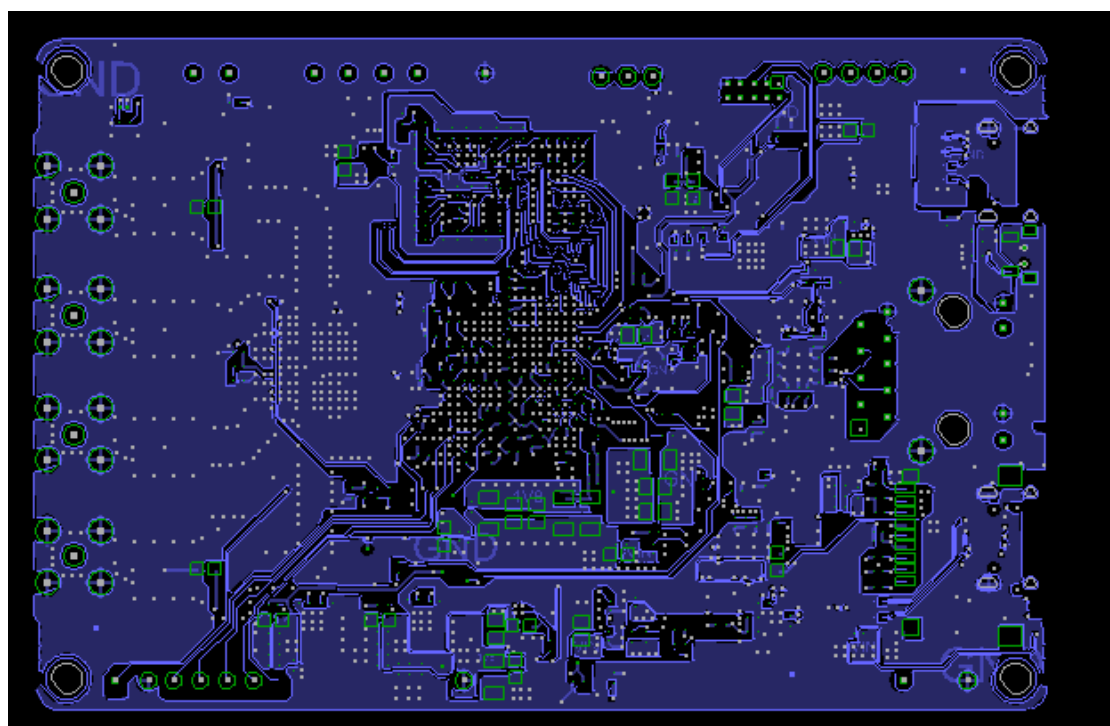
Impedance control information for this product:

IMPEDANCE TABLE				
NOTE: ALL IMPEDANCE TOLERANCE +/-10%				
SE = SINGLE ENDED, DP = DIFFERENTIAL PAIR				
LAYER	IMPEDANCE	REFERENCE	LINE WIDTH	SPACE
L1	50 OHMS SE	L2	4MILS	N/A
L1	50 OHMS SE	L4	19MILS	N/A
L1	40 OHMS SE	L2	6.5MILS	N/A
L1	100 OHMS DP	L4	9.1MILS	7MILS
L1	100 OHMS DP	L2	4.1MILS	7.9MILS
L1	90 OHMS DP	L2	5.1MILS	6MILS
L5	40 OHMS SE	L4,L7	6.5MILS	N/A
L5	50 OHMS SE	L4,L7	4.5MILS	N/A
L5	90 OHMS DP	L4,L7	5.1MILS	6MILS
L5	80 OHMS DP	L4,L7	5.5MILS	7MILS
L6	40 OHMS SE	L4,L7	6.5MILS	N/A
L6	50 OHMS SE	L4,L7	4.5MILS	N/A
L10	50 OHMS SE	L9	5MILS	N/A
L10	100 OHMS DP	L9	4MILS	8MILS
L10	40 OHMS SE	L9	6.5MILS	N/A
L10	80 OHMS DP	L9	5MILS	6MILS

The differential balun of the RF antenna adopts interlayer reference to increase the stability of the RF circuit. DDR signals, gigabit network port signals, and key control signals are all included in the impedance control category to increase system stability。

Screenshot of key circuit wiring for the product:





Screenshot of the product length control table section:

cadence

Objects		Referenced Electrical CSet	Pin Pairs	Pin Delay		Scope	Relative Delay				Length mil
Type	Name			Pin 1 mil	Pin 2 mil		Delta: Tolerance ns	Actual	Margin	+/-	
Dsn	Iomia										
MGrp	A-TX (8)		All Drivers/All Receiv...			Global	0.00 MIL: 10.00 MIL		1e-005 ns		
MGrp	ADDR/CTL TOCLK (26)		All Drivers/All Receiv...			Global	0 ns: 0.01 ns		1e-005 ns		
MGrp	CTRL_IN (4)		All Drivers/All Receiv...			Global	0.00 MIL: 500.00 MIL		392.02 MIL		
MGrp	CTRL_OUT (8)		All Drivers/All Receiv...			Global	0.00 MIL: 500.00 MIL		5.03 MIL		
MGrp	DDR3_BYTE_LANE_0 (11)		All Drivers/All Receiv...			Global	0 ns: 0.001 ns		0.00034 ns		
PPr	U2.E7:U4.A1 [DDR3_DM0]			0.0966...		Global	0 ns: 0.001 ns	0.00062 ns	0.00038 ns	-	1620.08
PPr	U4.B2:U2.G3 [DDR3_DQS0_N]			0.1013...		Global	0 ns: 0.001 ns	0.00013 ns	0.00087 ns	-	1632.98
PPr	U4.C2:U2.F3 [DDR3_DQS0_P]			0.1011...		Global	0 ns: 0.001 ns	7e-005 ns	0.00093 ns	-	1633.61
PPr	U2.H8:U4.C3 [DDR3_DQ0]			0.0953...		Global	0 ns: 0.001 ns	0.00045 ns	0.00055 ns	+	1628.69
PPr	U2.F7:U4.B3 [DDR3_DQ1]			0.1072...		Global	0 ns: 0.001 ns	0.00049 ns	0.00051 ns	-	1611.10
PPr	U2.F8:U4.A2 [DDR3_DQ2]			0.0944...		Global	0 ns: 0.001 ns	0.00024 ns	0.00076 ns	+	1628.13
PPr	U2.E3:U4.A4 [DDR3_DQ3]			0.1145...		Global	0 ns: 0.001 ns	TARGET			1607.86
PPr	U2.F2:U4.D3 [DDR3_DQ4]			0.0959...		Global	0 ns: 0.001 ns	0.00056 ns	0.00034 ns	-	1620.41
PPr	U2.H3:U4.D1 [DDR3_DQ5]			0.0952...		Global	0 ns: 0.001 ns	0.00049 ns	0.00051 ns	-	1622.25
PPr	U2.H7:U4.C1 [DDR3_DQ6]			0.1053...		Global	0 ns: 0.001 ns	0.00061 ns	0.00039 ns	-	1612.07
PPr	U2.G2:U4.E1 [DDR3_DQ7]			0.0966...		Global	0 ns: 0.001 ns	6e-005 ns	0.00094 ns	+	1624.82
MGrp	DDR3_BYTE_LANE_1 (11)		All Drivers/All Receiv...			Global	0 ns: 0.001 ns		0.00039 ns		
PPr	U2.D3:U4.F1 [DDR3_DM1]			0.0736...		Global	0 ns: 0.001 ns	0.0006 ns	0.0004 ns	+	1414.75
PPr	U4.F2:U2.B7 [DDR3_DQS1_N]			0.0786...		Global	0 ns: 0.001 ns	0.00032 ns	0.00068 ns	-	1383.55
PPr	U4.G2:U2.C7 [DDR3_DQS1_P]			0.0803...		Global	0 ns: 0.001 ns	0.00031 ns	0.00069 ns	-	1383.18
PPr	U2.D7:U4.E2 [DDR3_DQ8]			0.0715...		Global	0 ns: 0.001 ns	0.00018 ns	0.00082 ns	-	1383.52
PPr	U2.D7:U4.E3 [DDR3_DQ9]			0.0805...		Global	TARGET				1390.26
PPr	U2.A3:U4.G3 [DDR3_DQ10]			0.0646...		Global	0 ns: 0.001 ns	4e-005 ns	0.00096 ns	-	1386.01
PPr	U2.C8:U4.H3 [DDR3_DQ11]			0.0606...		Global	0 ns: 0.001 ns	8e-005 ns	0.00092 ns	-	1393.04
PPr	U2.A2:U4.J3 [DDR3_DQ12]			0.0626...		Global	0 ns: 0.001 ns	0.00058 ns	0.00042 ns	-	1383.10
PPr	U2.C3:U4.H2 [DDR3_DQ13]			0.0627...		Global	0 ns: 0.001 ns	0.00061 ns	0.00039 ns	-	1382.88
PPr	U4.H1:U2.B8 [DDR3_DQ14]			0.0669...		Global	0 ns: 0.001 ns	0.00029 ns	0.00071 ns	+	1442.10
PPr	U2.C2:U4.J1 [DDR3_DQ15]			0.0613...		Global	0 ns: 0.001 ns	9e-005 ns	0.00091 ns	-	1386.46
MGrp	ETH_RX (6)		All Drivers/All Receiv...			Global	0.00 MIL: 10.00 MIL		3.16 MIL		
PPr	U1.13:U4.D13 [ETH_RXCTL]			0.1408...		Global	0.00 MIL: 10.00 MIL	TARGET			2324.52
PPr	U1.19:U4.B17 [ETH_RXC]			0.1518...		Global	0.00 MIL: 10.00 MIL	5.28 MIL	4.72 MIL	-	2319.25
PPr	U1.14:U4.D11 [ETH_RXD0]			0.1099...		Global	0.00 MIL: 10.00 MIL	1.45 MIL	8.55 MIL	-	2323.08
PPr	U1.16:U4.A16 [ETH_RXD1]			0.1370...		Global	0.00 MIL: 10.00 MIL	3.55 MIL	6.45 MIL	+	2328.08
PPr	U1.17:U4.F15 [ETH_RXD2]			0.1341...		Global	0.00 MIL: 10.00 MIL	6.84 MIL	3.16 MIL	-	2317.69
PPr	U1.18:U4.A15 [ETH_RXD3]			0.1359...		Global	0.00 MIL: 10.00 MIL	3.19 MIL	6.81 MIL	+	2327.71
MGrp	ETH_TX (6)		All Drivers/All Receiv...			Global	0.00 MIL: 10.00 MIL		1.73 MIL		

Relative Propagation Delay/

DB SYNCHN

In Cadence, all DDR network ports and other wiring meet the control rules.

Supported firmware types

Fully compatible with ADI's official Pluto firmware, and supports third-party derivative firmware such as Plutoplus firmware in the market.

